

Real Estate Market Insights: An Exploratory Analysis of Zameen.com Listings in Pakistan

Project Objective:

To extract actionable insights from property listings on Zameen.com -such as pricing trends, neighborhood comparisons, and listing quality -that can help real estate investors make informed decisions.

Deliverables:

1. A Colab/ Jupyter Notebook with end-to-end analysis
2. An executive report or PDF summarizing key findings and visualizations

Rubric:

Code with comments: 50

Report: 50

Dataset Link and Description:

[Dataset link](#)

Instead of importing the whole dataset into pandas, keep in mind which columns are important to import and only import those (it is possible to only import certain columns while reading the csv into pandas) for example:

- City
- Location
- Price
- Property type
- Area
- Number of beds/baths/kitchen etc
- Description text

Project Sections

1. Problem Statement

- Define the business question: What drives property prices in Pakistan?

2. Data Understanding & Preprocessing

- Use: `pandas`, `numpy`
- Tasks:
 - View basic info
 - Handle duplicates

3. Missing Values Treatment

- Tasks:

- Identify missing values in whole data and in each column
- Impute with mode/mean/forward-fill for categorical/numerical data
- Justify treatment of missing values

4. Data Cleaning & Consistency

- Use: **FuzzyWuzzy**, **pandas** and user defined functions
- Tasks:
 - Standardize city names using Fuzzy Matching
 - Clean messy columns (e.g., remove commas, PKR symbols)
 - Convert area units (marla/kanal/sqft → uniform sqft), if required
 - Correct different categories in the purpose column
 - Detect and handle any outliers in prices that don't make sense

5. Feature Engineering

- Create new features/columns to derive more insightful analysis

6. Univariate & Bivariate Analysis

- Use: **matplotlib**, **seaborn**, **plotly**
- Tasks:
 - Histograms of numerical columns like prices, area etc and countplots of categorical columns like property type, locations etc
 - Different barplots of aggregated data

- Correlation heatmap of appropriate columns
- Violin plots / box plots to compare prices by city, property type, bedrooms

7. Insights & Recommendations

- Use: domain logic + stats

8. Conclusion & Next Steps

- Summary of learnings
- Suggestions for stakeholders (investors)